# FEDERICA ADOBBATI

Postdoc

## CONTACT


+39 3474265565

fe.adobbati@gmail.com

Liège, Belgium


## EDUCATION

**Oct 2016 - Oct 2018**

**UNIVERSITY OF MILANO - BICOCCA**

- Master in Computer Science
- Final grade: 110/110 cum laude

**Oct 2012 - Sep 2016**

**UNIVERSITY OF MILANO - BICOCCA**

- Bachelor in Mathematics
- Final grade: 95 / 110

## SKILLS

- Teamwork
- Scientific writing
- Science communication

## PROGRAMMING

- Python
- Java
- C

## LANGUAGES

- Italian
- English
- French
- German
- Dutch

## PROFILE

I am a researcher and a computer scientist. In my current job I develop deep learning algorithms to address oceanographic problems. During my Master and PhD in computer science I focused on formal analysis of complex system and co-supervised several bachelor and master theses.

## RESEARCH EXPERIENCE

**University of Liège**

Postdoc postition — 2025 - PRESENT

- Data processing and harmonization
- Development of generative deep learnin algorithms
- Modelling oxygen in the global ocean

**OGS, Institute of oceanography and applied geophysics**

Postdoc postition — 2023 - 2025

- Development of deep learning methods on HPC infrastructure
- Datadriven improvement of oceanographic data resolution
- Modelling biogeochemical properties of the northern Adriatic Sea

**University of Milano - Bicocca**

PhD student — 2019 - 2023

- Title: Formal verification of control and security properties on partially observable distributed systems
- Development of algorithms based on Petri nets
- Complexity analysis of algorithms

**University of Milano-Bicocca**

Post scholarship on the project FOODESART — 2018- 2019

- Modelling cooking recepies with Petri Nets
- Automatial representation of processes to enable formal analysis and optimization

## TEACHING AND SUPERVISION

**OGS**

- School “Digital Twin for sustainable economy ”, talk inside the day “Digital Twins of Marine Environments” - Oct 2024

**University of Milano-Bicocca**

- Participation in the project for high school pupils: “Playing seriously: models to understand concurrent systems” - Feb 2024
- Tutor of the courses:
  - “Models and Computations ” (2019-2021)
  - “Languages and computability” (2018/2019 - 2021/2022)
  - “Cybersecurity for datascience” (2020/2021)
- Tutor for student with disability in the course “General computer science” (2022/2023)
- Co-supervisor of several Bachelor and Master theses (2018-2024)

## SELECTION RECENT PUBLICATIONS

- A deep learning approach for coastal downscaling: the northern Adriatic Sea case-study - *Federica Adobbati, Lorenzo Bonin, Stefano Querin, Fabio Giordano, Valeria Di Biagio, Luca Manzoni, Gianpiero Cossarini* (Ocean Modeling)
- A two player asynchronous game with privacy constraints on Petri nets - *Federica Adobbati, Luca Bernardinello, Lucia Pomello* (ICTCS24)
- Computing a parametric reveals relation for bounded equal-conflict Petri nets - *Federica Adobbati, Luca Bernardinello, Gorkem Kılınç Soylu, Lucia Pomello* (Transactions on Petri Nets and Other Models of Concurrency XVII, 54-83 - 2023)
- A finite prefix for analysing information flow among transitions of a free-choice net - *Federica Adobbati, Gorkem Kılınç Soylu, Adrian Puerto Aubel* (IEEE Access 10, 38483-38501 - 2022)
- Analysing multi-agent systems using 1-safe Petri nets - *Federica Adobbati, Lukasz Mikulski* - arXiv preprint 2023

## RECENT TALKS AND POSTER

- NOAA coastal ocean modelling seminar "Deep learning methods for super-resolution tasks in coastal ocean modelling: the Northern Adriatic Sea case-study" - August 2025
- Talk at COSS-TT "A deep learning approach for coastal downscaling: the northern Adriatic Sea case-study" - June 2025
- Talk and poster at the "International Workshop on Digital Twins" - Title: "Deep learning methods for super-resolution tasks in ocean modelling" - Feb 2025
- RETURN Dissemination Workshop "Multi risk models from assessment to validation and implication for decision making" - Poster "Hazard indicators for marine ecosystems" - Jun 2024
- Talk at iNE(S)T-working on deep learning methods for the super-resolution of the northern Adriatic Sea - Feb 2024